KAIZEN MATHS

Lesson starter

Quadratic Equations

Square-Root Method

Name _____

S1

$x^2 - 9 = 0$

S2

$x^2 - 5 = -1$

S3

$x^2 - 36 = 0$

S4

$x^2 - 12 = 13$

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E1

$x^2 - 25 = 0$

E2

$x^2 - 11 = 38$

E3

$x^2 - 16 = 0$

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Teacher answers & working

Quadratic Equations · Square-Root Method

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S1

$x^2 - 9 = 0$

Answer:

$x = +/-3$

Working:

Step 1: Use the square-root method: isolate x^2, then take both roots.

Step 2: Add 9 to both sides:

$x^2 = 9$

Step 3: Take the square root of both sides:

sqrt(x^2) = +/-sqrt(9)

$x = +/-3$

Step 4: Final answer: x = +/-3

S2

$x^2 - 5 = -1$

Answer:

$x = +/-2$

Working:

Step 1: Use inverse operations to isolate x^2 before taking square roots.

Step 2: Add 5 to both sides:

$x^2 = 4$

$x^2 = 4$

Step 3: Take the square root of both sides:

$x = +/-2$

Step 4: Final answer: x = +/-2

Teacher answers & working

Quadratic Equations · Square-Root Method

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S3

$x^2 - 36 = 0$

Answer:

$x = +/-6$

Working:

Step 1: Use the square-root method: isolate x^2, then take both roots.

Step 2: Add 36 to both sides:

$x^2 = 36$

Step 3: Take the square root of both sides:

sqrt(x^2) = +/-sqrt(36)

$x = +/-6$

Step 4: Final answer: x = +/-6

S4

$x^2 - 12 = 13$

Answer:

$x = +/-5$

Working:

Step 1: Use inverse operations to isolate x^2 before taking square roots.

Step 2: Add 12 to both sides:

$x^2 = 25$

$x^2 = 25$

Step 3: Take the square root of both sides:

$x = +/-5$

Step 4: Final answer: x = +/-5

Teacher answers & working

Quadratic Equations · Square-Root Method

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E1

$x^2 - 25 = 0$

Answer:

$x = +/-5$

Working:

Step 1: Use the square-root method: isolate x^2, then take both roots.

Step 2: Add 25 to both sides:

$x^2 = 25$

Step 3: Take the square root of both sides:

sqrt(x^2) = +/-sqrt(25)

$x = +/-5$

Step 4: Final answer: x = +/-5

E2

$x^2 - 11 = 38$

Answer:

$x = +/-7$

Working:

Step 1: Use inverse operations to isolate x^2 before taking square roots.

Step 2: Add 11 to both sides:

$x^2 = 49$

$x^2 = 49$

Step 3: Take the square root of both sides:

$x = +/-7$

Step 4: Final answer: x = +/-7

Teacher answers & working

Quadratic Equations · Square-Root Method

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E3

$x^2 - 16 = 0$

Answer:

$x = +/-4$

Working:

Step 1: Use the square-root method: isolate x^2, then take both roots.

Step 2: Add 16 to both sides:

$x^2 = 16$

Step 3: Take the square root of both sides:

sqrt(x^2) = +/-sqrt(16)

$x = +/-4$

Step 4: Final answer: x = +/-4